

4-2 The Mean Value Theorem

Homework : 1,3,5,11,13,15,17,23,25,27,29,35

Rolle's Theorem

Assumptions : $\left. \begin{array}{l} (1) f \text{ is continuous on } [a, b] \\ (2) f \text{ is differentiable on } (a, b) \end{array} \right\} \text{簡稱 } f \text{ is smooth on } [a, b]$
(3) $f(a) = f(b)$ (等高)

$$\Rightarrow \exists c \in (a, b) \text{ s.t. } f'(c) = 0$$

● Smooth (平滑) + 等高

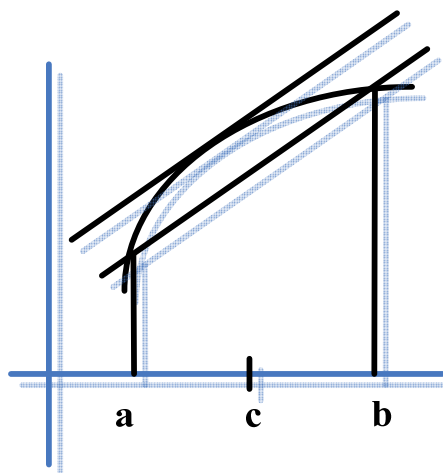
$\Rightarrow \begin{cases} 1. \text{數學: 有切線斜率為 } 0 \text{ 的點(即有峰或谷)} \\ 2. \text{物理: 有速度為 } 0 \text{ 的時間} \end{cases}$

Mean Value Theorem (MVT)

(1) + (2) (即 f is smooth on $[a, b]$)

$$\Rightarrow \exists c \in (a, b) \text{ s.t.}$$

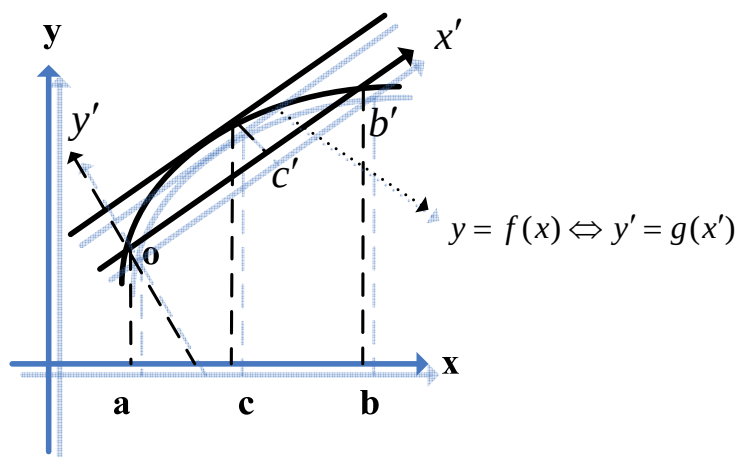
$$f(b) - f(a) = f'(c)(b - a)$$



$$\frac{f(b) - f(a)}{b - a} = \text{割線斜率}$$
$$f'(c) = \text{切線斜率}$$

若 f 在 $[a, b]$ 是平滑的 $\Rightarrow \begin{cases} \text{數學上: 割線斜率} = \text{某一切線斜率} \\ \text{物理上: 平均速度} = \text{某一瞬間速度} \end{cases}$

Proof of MVT :



Proof : 在 $x' - y'$ 新座標系，將 Rolle 用在 $y' = g(x')$

$$\Rightarrow \exists c' \in (0, b') \text{ s.t. } g'(c') \parallel x' \text{ 軸}$$

$$\Leftrightarrow \exists c \in (a, b) \text{ s.t. } f'(c) \parallel x' \text{ 軸}$$

$$\Leftrightarrow f'(c) = \frac{f(b) - f(a)}{b - a}$$

Example 1 : $f(x) = x^3 - x$, $[a, b] = [0, 2]$, Find c .

$$\text{Solution : } \frac{6 - 0}{2 - 0} = 3c^2 - 1 \Rightarrow c = \frac{2\sqrt{3}}{3}.$$

Example 2 : f : 平滑 $f(0) = -3$, $f'(x) \leq 5$, $0 \leq x \leq 2$.

How large can $f(2)$ possibly be?

$$\text{Solution : } \frac{f(2) + 3}{2 - 0} = f'(c) \leq 5 \Rightarrow f(2) \leq 7 \Rightarrow f(2) \text{ 最大的可能是 } 7.$$

Example 3 : f : 平滑 $f(0) = -3$, $f'(x) \geq 2$, $0 \leq x \leq 4$.

How small can $f(4)$ possibly be?

$$\text{Solution : } \frac{f(4) + 3}{4 - 0} = f'(c) \geq 2 \Rightarrow f(4) \geq 5 \Rightarrow f(4) \text{ 最小的可能是 } 5.$$

Example 4 : Does there exist a function f s.t.

$$f(0) = -1, f(2) = 4, \text{ and } f'(x) \leq 2 \text{ for all } x?$$

Solution : $\frac{4+1}{2-0} = \frac{5}{2} = f'(c) \Rightarrow$ 不可能.

Example 5 : Prove that $\sqrt{1+x} < 1 + \frac{1}{2}x$ if $x > 0$.

Solution : Let $f(x) = (1+x)^{\frac{1}{2}} \Rightarrow f'(x) = \frac{1}{2}(1+x)^{-\frac{1}{2}}$

$$\stackrel{MVT}{\Rightarrow} f(x) - f(0) = f'(c)(x-0), c \in (0, x)$$

$$\Rightarrow \sqrt{1+x} - 1 = \frac{1}{2}(1+c)^{-\frac{1}{2}}x.$$

$$\Rightarrow \sqrt{1+x} = 1 + \frac{1}{2}(1+c)^{-\frac{1}{2}}x < 1 + \frac{1}{2}x.$$

$$(\because (1+c)^{-\frac{1}{2}} < 1)$$

Example 6 : Prove that $|\cos a - \cos b| \leq |a - b|$ for all a and b .

Proof : Let $f(x) = \cos x$, assume without loss of generality, that $a \geq b$

$$\stackrel{MVT}{\Rightarrow} \cos a - \cos b = f'(c)(a-b) = (\sin c)(a-b)$$

$$\Rightarrow |\cos a - \cos b| \leq |a-b|.$$

$$(\because |\sin c| \leq 1).$$

Example 7 : $f'(x) = 0$ on $(a, b) \Rightarrow f$ is constant on (a, b) .

Proof : Let $x_1, x_2 \in (a, b), x_1 < x_2$

$$\Rightarrow f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0, \quad c \in (x_1, x_2)$$

$$\Rightarrow f(x_2) = f(x_1)$$

$$\Rightarrow f \text{ has the same value at any two numbers } x_1, x_2 \in (a, b)$$

$$\Rightarrow f \text{ is constant on } (a, b).$$

Example 8 : $f'(x) = g'(x)$ for all $x \in (a, b)$

$$\Rightarrow f(x) = g(x) + c \text{ for all } x \in (a, b).$$

Here c is a constant.

Proof : A direct consequence of Ex7.